XenMange

Administrator & User Manual

A modern, futuristic web interface for administering XenServer pools, hosts, VMs, storage, and networking.

Version 1.0.0

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Part I — Introduction

1. Product Overview

XenMange is a modern, web-based management interface for **Citrix XenServer** (and XCP-ng) virtualization infrastructure. It replaces the legacy XenCenter desktop application with a responsive, browser-based experience that supports managing pools, hosts, virtual machines, storage repositories, and virtual networks — all from a single pane of glass.

XenMange communicates with XenServer via the **XenAPI JSON-RPC v2.0** protocol over HTTPS, and persists all configuration, credentials, telemetry, and audit data in local **SQLite** databases.

Key Capabilities

- **Full VM Lifecycle:** Create, start, shutdown, reboot, suspend, resume, clone, copy, migrate, snapshot, import, and export virtual machines.
- **Multi-Pool Management:** Attach to and switch between multiple XenServer pools from a single session.
- **Host Operations:** Enter/exit maintenance mode, reboot, shutdown, and monitor host health.
- **Storage & Networking:** View and manage storage repositories and virtual networks across pools.
- **Real-Time Telemetry:** Background metrics collection for hosts, VMs, and storage with hourly rollups and configurable retention.
- **Alert Center:** Persisted alerts with acknowledgement, suppression, severity overrides, and remediation task creation.
- **Governance & RBAC:** Role-based access control (read-only, operator, admin), destructive-action approval workflows, pool quotas, and audit trails.
- **Credential Vault:** AES-256-GCM envelope encryption for stored XenServer passwords with key rotation support.
- **Template Governance:** Version tracking, golden-image baselines, compare-and-promote workflows.
- **Resilience & HA/DR:** Protection coverage tracking, failover readiness, evacuation targets, recovery runbooks and drills.
- **Log Export:** Federated log center with JSON, HTML, and PDF export capabilities.

2. Design Philosophy & UI Language

XenMange adopts a **"post-modern futuristic dark glassmorphic"** design language — described internally as *"Matrix meets Hackers"*. The interface features:

- **Dark glassmorphic theme:** Translucent panels with backdrop blur effects over a dark background.
- **Neon green accent palette:** Matrix-green primary color (#0a0) for highlights, borders, and interactive elements.

- **Scanline animations:** Subtle CRT-style scanline overlays on key surfaces.
- **Floating windows:** Dialogs and forms appear as floating glassmorphic panels.
- **Dense tree navigation:** Side navigation with collapsible sections for efficient workspace switching.
- **Material Design Icons:** @mdi/font icon set for consistent iconography throughout.
- **Monospace data displays:** Share Tech Mono and Rajdhani fonts for technical data presentation.

3. System Requirements

Server Requirements

Component	Minimum	Recommended
Operating System	Linux (any recent distro), macOS, Windows	Ubuntu 22.04+ / Debian 12+
Node.js	v18.0+	v20 LTS or v22 LTS
RAM	512 MB	1 GB+
Disk	200 MB (application + initial databases)	2 GB+ (telemetry storage depends on retention)
CPU	1 core	2+ cores
Network	HTTPS access to XenServer hosts (port 443)	Low-latency LAN connection to XenServer management interface

Build Requirements

- **C++ build tools** (gcc/g++/make) — required to compile the `better-sqlite3` native addon.
- **npm** — for installing dependencies and running scripts.

Client Requirements

- Any modern browser: Chrome 90+, Firefox 90+, Edge 90+, Safari 15+.
- JavaScript must be enabled.

XenServer Requirements

- XenServer 7.x+ or XCP-ng 8.x+ with JSON-RPC API enabled.
- HTTPS access on port 443.
- A user account with sufficient privileges (typically `root`).

Part II — Architecture

4. High-Level Architecture

XenMange follows a three-tier architecture:

```
┌─────────────────────────────────────────────────────────────┐
│                    Browser (Client)                         │
│           Vue.js 3 SPA + Vue Router 4                       │
│      Pre-compiled templates (CSP-safe)                      │
│      Material Design Icons (local)                          │
└──────────────────────┬──────────────────────────────────────┘
                       │ HTTP (session cookie)
┌──────────────────────▼──────────────────────────────────────┐
│                 Express 5 Server                            │
│                                                             │
│   ┌─────────────────────────────────────────────────────┐   │
│   │ Middleware: Helmet (CSP) → CORS → Rate Limit →      │   │
│   │ Session → Validation (Joi) → Governance (RBAC)      │   │
│   └─────────────────────────────────────────────────────┘   │
│                                                             │
│   ┌───────────┐  ┌──────────────┐  ┌───────────────────┐    │
│   │  Routes   │  │   Services   │  │   Models (DB)     │    │
│   │ (22 files)│  │  (20 files)  │  │  (4 SQLite DBs)   │    │
│   └───────────┘  └──────┬───────┘  └───────────────────┘    │
└─────────────────────────┼───────────────────────────────────┘
                          │ HTTPS (JSON-RPC v2.0)
             ┌────────────▼────────────┐
             │   XenServer / XCP-ng    │
             │   (Pool of Hosts)       │
             └─────────────────────────┘
```

Request Flow

1. Browser sends HTTP request with session cookie.
2. Express middleware pipeline processes the request: security headers → CORS → rate limiting → session restoration → input validation.
3. Route handler executes, potentially invoking service-layer logic.
4. If a XenServer connection is required, the service calls `xenapi.js` which sends JSON-RPC to XenServer over HTTPS.
5. Response is returned to the browser as JSON.

5. Technology Stack

Layer	Technology	Purpose
Runtime	Node.js (CommonJS)	Server-side JavaScript execution
Web Framework	Express 5.2	HTTP routing, middleware, static serving
Frontend Framework	Vue.js 3.5 (SPA)	Reactive UI components
Client Router	Vue Router 4.5	Client-side URL routing
Templating	EJS 6	Server-rendered app shell + error pages
Database	SQLite via better-sqlite3 13	Local persistence (4 separate DB files)
Authentication	bcryptjs + express-session	Password hashing + session management
Validation	Joi 18	Request body/query parameter validation
Security Headers	Helmet 8	CSP, X-Frame-Options, nosniff, etc.
Encryption	Node.js crypto (AES-256-GCM)	Credential vault envelope encryption
PDF Generation	pdfkit 0.20	Log export to PDF
HTTP Client	axios 1.19	XenServer JSON-RPC communication
Icons	@mdi/font 7.4	Material Design Icons (local bundle)
Unit Testing	Jest 30	Server-side unit tests
E2E Testing	Playwright 1.62	Browser-based end-to-end tests

> **Note:** All frontend libraries (Vue, Vue Router, Material Design Icons) are bundled locally — no CDN dependencies. This ensures full CSP compliance and offline capability.

6. Database Architecture

XenMange uses **four separate SQLite database files**, each serving a distinct domain. This isolation prevents lock contention, simplifies backups, and allows independent retention policies.

Database File	Purpose	Key Tables
`xenmange.db`	Main application data	`connections`, `host_targets`, `settings`, `retention_policies`, `deployment_runs`, `deployment_run_steps`
`security.db`	Users, groups, sessions, auth events, vault key material	`users`, `groups`, `group_members`, `sessions`, `auth_events`, `vault_key_material`
`vault.db`	Encrypted credential blobs (ciphertext only)	`credentials`
`perf.db`	Time-series telemetry data	`metric_samples`, `metric_hourly_rollups`

Session Persistence

Express sessions are stored in the `sessions` table within `security.db`. Sessions survive server restarts — when the server restarts, it rehydrates XenServer API connections from persisted session references.

Schema Migrations

All database modules use `CREATE TABLE IF NOT EXISTS` for initial setup and perform runtime migration by checking `PRAGMA table_info()` for missing columns, adding them via `ALTER TABLE` as needed. No separate migration tool is required.

7. Security Architecture

Authentication Layers

XenMange has **two independent authentication systems**:

1. **Control-Plane Login:** Authenticates against the local `users` table in `security.db` using bcrypt-hashed passwords. This grants access to the XenMange application itself.
2. **XenServer Login:** Authenticates against XenServer using JSON-RPC session creation. This is separate from the control-plane and can target multiple XenServer instances simultaneously.

Session Management

- Cookie name: `xenmange.sid`
- HttpOnly, SameSite=Lax
- Secure flag enabled in production
- Default max age: 24 hours (configurable)
- SQLite-backed persistence (survives restarts)

Credential Vault

Stored XenServer passwords are encrypted using **AES-256-GCM** with envelope encryption:

- A **master key** (from `VAULT_ENCRYPTION_KEY` env var) wraps per-credential Data Encryption Keys (DEKs).
- DEKs encrypt the actual passwords.
- Wrapped DEKs are stored in `security.db`; encrypted passwords are in `vault.db`.
- Plaintext passwords are **never** returned to the browser.
- Key rotation is supported via a previous-key fallback mechanism.

Security Headers (Helmet)

Header	Value
Content-Security-Policy	default-src 'self'; script-src 'self' 'unsafe-inline'; style-src 'self' 'unsafe-inline' https://fonts.googleapis.com; font-src 'self' https://fonts.gstatic.com; img-src 'self' data: https:; connect-src 'self'; frame-src 'none'; object-src 'none'
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
Referrer-Policy	strict-origin-when-cross-origin

Rate Limiting

Login endpoints (`/api/auth/login` and `/api/auth/xen-login`) are rate-limited to **20 attempts per 15 minutes per IP address**.

Input Validation

Over 60 Joi schemas validate every API endpoint. All inputs are validated with `stripUnknown: true` to prevent field injection, and `abortEarly: false` to report all errors at once.

8. API Architecture

Route Groups

Routes are organized into two authentication tiers:

Auth Tier	Description	Routes
Public	No authentication required	`/api/auth/login` , `/api/auth/xen-login` , static assets
requireAuth	Authenticated control-plane user (no Xen connection required)	`/api/audit` , `/api/governance` , `/api/settings` , `/api/logs` , `/api/credentials` , `/api/users` , `/api/groups` , `/api/connections` , `/api/host-targets` , `/api/workspaces`
requireXenConnection	Authenticated + active XenServer connection	`/api/dashboard` , `/api/vms` , `/api/hosts` , `/api/storage` , `/api/networks` , `/api/pools` , `/api/tasks` , `/api/resilience` , `/api/lifecycle` , `/api/alerts` , `/api/metrics`

Multi-Target Session Architecture

XenMange supports multiple simultaneous XenServer connections. Each connection is identified by a `targetKey` (e.g., `connection:1` or `host:10.0.0.1|user:root|port:443`). The session persists target descriptors, and the connection registry holds live XenAPI instances in memory.

Target selection priority:

1. Explicit `targetKey` in request body/query/header (`X-XenMange-Target-Key`)
2. Explicit `targetConnectionId` in request
3. Active target from session (`activeXenTargetKey`)
4. First available target

Part III — Installation & Configuration

9. Fresh Installation

Step 1: Clone the Repository

```
git clone <repository-url> XenMange
cd XenMange
```

Step 2: Install Dependencies

```
npm install
```

> **Note:** The `better-sqlite3` package requires C++ build tools. On Ubuntu/Debian: `sudo apt-get install build-essential python3`. On RHEL/CentOS: `sudo yum groupinstall "Development Tools"`.

Step 3: Configure Environment

```
cp .env.example .env
```

Edit `.env` with your production values (see Section 10 for the full reference).

Step 4: Build the Client

```
npm run build:client
```

This compiles Vue templates, bundles all JavaScript, and copies vendor assets into `client/dist/`.

Step 5: Start the Server

```
npm start
```

The server starts on port 3000 (default). Open `http://localhost:3000` in your browser.

Development Mode

```
npm run dev
```

Uses `nodemon` for auto-reload on file changes. Automatically rebuilds the client and restarts the server when source files are modified.

10. Environment Configuration Reference

Variable	Default	Description
`NODE_ENV`	`development`	Environment mode. Set to `production` for production deployments.
`PORT`	`3000`	HTTP port the Express server listens on.
`SESSION_SECRET`	`xenmange-dev-secret-change-me`	Secret used to sign session cookies. **Must** be changed in production.
`SESSION_MAX_AGE`	`86400000` (24h)	Session cookie lifetime in milliseconds.
`XENMANGE_BOOTSTRAP_USERNAME`	`admin`	Username for the initial admin account created on first boot.
`XENMANGE_BOOTSTRAP_PASSWORD`	`admin123!`	Password for the initial admin account. **Must** be changed in production.
`XENMANGE_BOOTSTRAP_DISPLAY_NAME`	`Platform Administrator`	Display name for the initial admin.
`DB_PATH`	`./data/xenmange.db`	Path to the main application database.
`SECURITY_DB_PATH`	`./data/security.db`	Path to the security database (users, sessions, auth events).
`VAULT_DB_PATH`	`./data/vault.db`	Path to the encrypted credential vault database.
`PERF_DB_PATH`	`./data/perf.db`	Path to the performance/telemetry database.
`VAULT_ENCRYPTION_KEY`	(empty)	32-byte base64-encoded master key for credential vault encryption. Required in production. In development, a key is derived from `SESSION_SECRET`.
`VAULT_ENCRYPTION_KEY_PREVIOUS`	(empty)	Previous master key for rotation window. When set, decryption tries the current key first, then falls back to this key.

Security Warning: Never commit `.env` to version control. It is listed in `.gitignore` by default.

11. Production Hardening

Reverse Proxy (Recommended)

XenMange listens on plain HTTP. For production, place a reverse proxy (nginx, Caddy, HAProxy) in front to handle:

- TLS termination (HTTPS)
- Rate limiting at the edge

- Request buffering
- Logging

Generate a Vault Encryption Key

```
node -e "console.log(require('crypto').randomBytes(32).toString('base64'))"
```

Set the output as `VAULT_ENCRYPTION_KEY` in your `.env` file.

Change Default Credentials

Immediately change the bootstrap admin password after first login. Use the Settings workspace or the `/api/users/:id/password` endpoint.

Generate a Strong Session Secret

```
node -e "console.log(require('crypto').randomBytes(48).toString('base64'))"
```

File Permissions

```
chmod 600 .env
chmod 700 data/
```

CORS Restriction

The default CORS configuration allows all origins. In production, configure your reverse proxy to restrict CORS to your actual domain.

> **Important:** XenMange is a single-process, single-server application. Running multiple instances against the same SQLite databases will cause data corruption.

Part IV — Getting Started

12. First Login & Bootstrap

1. Open your browser and navigate to `http://<server>:3000`.
2. You will see the XenMange login screen with a dark glassmorphic interface.
3. Enter the bootstrap credentials:
   - **Username:** `admin` (or the value of `XENMANGE_BOOTSTRAP_USERNAME`)
   - **Password:** `admin123!` (or the value of `XENMANGE_BOOTSTRAP_PASSWORD`)
4. Click **Sign In**. You will be redirected to the Dashboard.

> **Critical:** Change the default bootstrap password immediately after first login. Use the Governance workspace → User Management section.

13. Connecting to XenServer

After logging in, you need to connect to a XenServer pool or individual host:

1. Navigate to **Settings** → **Connections** (or use the connection prompt on first login).
2. Click **Add Connection**.
3. Enter the XenServer details:
   - **Host:** IP address or hostname of the XenServer (e.g., `10.0.0.10`)
   - **Port:** `443` (default)
   - **Username:** `root` (or a user with sufficient privileges)
   - **Password:** The XenServer root password (encrypted and stored in the vault)
4. Optionally enable **Save Credentials** to persist the password in the encrypted vault.
5. Click **Connect**.

Once connected, the Dashboard will populate with pool, host, VM, and storage data from your XenServer environment.

Multi-Target Connections

XenMange supports attaching to multiple XenServer pools simultaneously. After connecting to a first target, use the target selector in the navigation bar to switch between active connections, or add additional connections in Settings.

Credential-Based Login

Alternatively, you can use stored credentials from the Credential Vault:

1. Navigate to **Settings** → **Credential Vault**.

2. Save a credential with the XenServer host/username/password.
3. When connecting, select the stored credential instead of typing the password.

14. The Dashboard

The Dashboard is your landing page after login. It provides an at-a-glance summary of your XenServer environment:

- **Pool Summary:** Number of pools, total hosts, overall health status.
- **Host Summary:** Hosts online/offline/maintenance, aggregate CPU and memory utilization.
- **VM Summary:** Total VMs, running/stopped/paused counts, resource consumption.
- **Storage Summary:** Storage repositories, total/allocation/usage.
- **Network Summary:** Virtual networks count and status.
- **Recent Messages:** Latest XenServer event messages.

Summary cards are **draggable** — you can rearrange them to suit your preferred layout.

15. Navigation Overview

XenMange uses a three-part navigation system:

Top Navigation Bar

- **Logo/App Name:** Click to return to the Dashboard.
- **Target Selector:** Dropdown showing the active XenServer connection. Switch between multiple connected targets.
- **Role Indicator:** Shows your current session role (read-only, operator, admin).
- **User Menu:** Profile, settings, logout.

Side Navigation Panel

Organized into collapsible sections:

- **Infrastructure:** Dashboard, Pools, Hosts, VMs, Storage, Networking, Templates
- **Operations:** Alerts, Activity, Lifecycle, Capacity, Resilience
- **Administration:** Governance, Inventory, Settings

Status Bar

Bottom bar showing connection status, current time, active session info, and telemetry collector status.

Part V — Feature Guides

16. Virtual Machine Management

The VMs workspace provides a comprehensive view of all virtual machines across your connected XenServer pools.

Viewing VMs

The VMs list displays each VM with its name, status (Running/Stopped/Paused/Suspended), power state, CPU count, memory allocation, and host assignment. Use the search bar to filter VMs by name.

Power Operations

Operation	Description	Approval Required
Start	Powers on the VM	Yes (if governance requires)
Shutdown (Clean)	Graceful shutdown via guest OS	Yes
Shutdown (Hard)	Immediate power-off (equivalent to pulling the power cord)	Yes
Reboot (Clean)	Graceful reboot via guest OS	Yes
Reboot (Hard)	Immediate reset	Yes
Suspend	Pauses VM and saves state to disk	Yes
Resume	Restores a suspended VM	No

> **Note:** Destructive power operations (start, shutdown, reboot, suspend) require governance approval tokens when the governance policy has destructive-action approval enabled.

VM Configuration

Click a VM to view its detailed configuration including:

- General info (name, description, UUID, OS type)
- Virtual CPUs (count, topology)
- Memory (static max, dynamic max/min)
- Virtual disks (VDI references, sizes, SR placement)
- Virtual NICs (network, MAC address, device position)
- Console endpoints
- Snapshot history

VM Duplication

XenMange supports two VM duplication modes:

- **Fast Clone (VM.clone):** Creates a linked clone that shares the base disk. Fast but requires the original disk to remain available.
- **Full Copy (VM.copy):** Creates an independent copy with its own disks. Slower but fully self-contained.

VM Console

Click the **Console** button on any running VM to open a console session. XenMange enumerates XAPI console records and launches session-authenticated console endpoints.

17. Host Management

The Hosts workspace lists all XenServer hosts across connected pools.

Host Information

For each host, XenMange displays:

- Name, address, and UUID
- Software version (XenServer/XCP-ng release)
- CPU model, count, and utilization
- Total, used, and free memory
- Resident VMs count
- Live metrics (memory, CPU trends)

Maintenance Mode

Entering maintenance mode on a host safely evacuates running VMs before allowing host-level operations (reboot, shutdown, patching).

1. Select a host and click **Enter Maintenance Mode**.
2. Configure the maintenance form:
   - **Evacuation Network:** Network to use for live migration of VMs off this host.
   - **Batch Size:** Number of VMs to migrate concurrently.
3. Confirm the operation. XenMange will:
   - Live-migrate VMs off the host in batches.
   - Mark the host as in maintenance mode once all VMs are evacuated.

To exit maintenance mode, click **Exit Maintenance Mode**. VMs can then be migrated back.

Host Power Operations

Operation	Description
Reboot	Reboots the host. All running VMs should be migrated off first.
Shutdown	Powers off the host. All running VMs should be migrated off first.

18. Storage Management

The Storage workspace displays all Storage Repositories (SRs) across connected pools.

For each SR, XenMange shows:

- Name, type (NFS, iSCSI, local, etc.), and UUID
- Total capacity, allocated, and used space
- Utilization percentage
- Parent pool
- Virtual Disks (VDIs) contained in the SR

19. Networking

The Networking workspace displays all virtual networks across connected pools.

For each network, XenMange shows:

- Name, description, and UUID
- VLAN tag (if applicable)
- Bridge name
- Parent pool
- Connected VIFs (Virtual Interfaces)

20. Pool Management

The Pools workspace lists all XenServer pools across your connected targets.

For each pool, XenMange shows:

- Pool name and UUID
- Master host
- Number of hosts
- Default SR and guest metrics host
- HA configuration status

21. Template Management & Governance

XenMange provides a governed template management system for maintaining golden images.

Template Governance

The governance system allows you to:

- **Version Templates:** Assign version labels to templates for tracking.
- **Mark Golden Images:** Set a template as the baseline/golden image for its family.
- **Compare Versions:** Compare configuration differences between template versions.
- **Promote Templates:** Promote a newer template to the golden image baseline, deprecating the previous version.
- **Retire Old Baselines:** Mark previous golden images as retired.

Template Deployment

Deploy VMs from governed templates with validation:

1. Select a template and click **Deploy**.
2. Configure deployment parameters (name, host, storage, network).
3. Run deployment validation to check resource availability.
4. Execute the deployment. Each step is tracked in the deployment run history.

22. VM Snapshots & Checkpoints

XenMange provides full snapshot and checkpoint management.

Creating Snapshots

1. Select a VM and navigate to the **Snapshots** section.
2. Click **Create Snapshot**.
3. Provide a name and optional description.
4. Choose snapshot type:
    - **Snapshot:** Captures disk state only (VM must be halted).
    - **Checkpoint:** Captures both disk and memory state (VM must be running).

Reverting to a Snapshot

Click **Revert** on any snapshot to restore the VM to that point. This is a **destructive action** that overwrites the current VM state and requires governance approval.

Deleting Snapshots

Click **Delete** on any snapshot to permanently remove it. This requires governance approval.

23. VM Migration

XenMange supports two migration modes:

Same-Pool Live Migration

Migrates a VM from one host to another within the same pool:

1. Select a VM and click **Migrate**.
2. Select the destination host from the pool.
3. Confirm the migration. The VM will be live-migrated with minimal downtime.

Cross-Pool Migration

Migrates a VM from one pool to an entirely different pool:

1. Select a VM and click **Migrate**.
2. Choose **Cross-Pool Migration**.
3. Configure:
   - **Destination Pool:** The target pool.
   - **Transfer Network:** Network for data transfer between pools.
   - **Target SR:** Storage repository on the destination for VM disks.
   - **Network Remapping:** Map each source VIF to a destination network.
4. Confirm the migration.

VM Compatibility

Before migration, XenMange checks host compatibility using `VM.get_possible_hosts` and `VM.assert_can_boot_here`. A compatibility matrix shows which hosts can run the VM, with CPU family alignment hints.

24. VM Import & Export

Exporting VMs

1. Select a VM and click **Export**.
2. Choose export format:
   - **XVA:** Full VM export including configuration and disks. XenServer-native format.
   - **Metadata Only:** Exports VM configuration without disk data.
3. The export file will be downloaded to your browser.

Importing VMs

1. Navigate to VMs → **Import**.

2. Select the import file (XVA or metadata archive).
3. Configure:
    - **Target Host:** Destination host for the import.
    - **Target SR:** Storage repository for the VM disks.
4. Start the import. Progress is tracked in the UI.

25. Alerts Center

The Alerts Center aggregates all alerts from XenServer and internal telemetry thresholds.

Alert Sources

- **XenServer Alerts:** Native XenServer alert messages (host down, SR full, etc.).
- **Telemetry Alerts:** Generated by XenMange's metrics collector when thresholds are exceeded (e.g., host memory > 85%).

Alert Management

Action	Description
Acknowledge	Mark an alert as reviewed. Removes it from the active alert count.
Suppress	Hide an alert from the active view without acknowledging it.
Override Severity	Manually change the severity level of an alert.
Bulk Triage	Select multiple alerts and acknowledge/suppress them at once.
Deep Link	Navigate directly to the affected object (host, VM, SR).
Create Remediation Task	Generate a remediation task from an alert for tracking resolution.

Alert Policies

Configure alert policies to control which alerts are generated and at what thresholds. Policies can be created, updated, and deleted (with governance approval for deletion).

26. Activity & Log Center

The Activity workspace provides a federated view of all operational events across the system.

Log Sources

- **Audit Log:** Operator actions and configuration changes.
- **Auth Events:** Login/logout events for all users.
- **Alert Records:** Alert state changes (created, acknowledged, suppressed).
- **Remediation Tasks:** Task creation and status updates.

- **Xen Background Tasks:** Background operations from XenServer.
- **Telemetry Events:** Metrics collection and threshold violations.

Export Options

Format	Description
JSON	Machine-readable export of all log entries.
HTML	Formatted HTML report with print-friendly styling.
PDF	Professional PDF document generated via pdfkit.

27. Metrics & Telemetry

XenMange includes a built-in background metrics collector that polls connected XenServer targets on a configurable schedule.

Metrics Collected

Entity	Metrics
Host	`memory_total_bytes`, `memory_free_bytes`, `memory_used_bytes`, `memory_used_percent`
VM	`memory_actual_bytes`, `memory_static_max_bytes`, `memory_usage_percent`, `vcpu_count`
Storage (SR)	`allocation_bytes`, `physical_bytes`, `utilization_percent`

Time Ranges

Metrics are available in five time ranges: `1h`, `6h`, `24h`, `7d`, and `30d`. For the 7d and 30d ranges, XenMange automatically uses hourly rollups instead of raw samples for efficient querying.

Collection Schedule

- **Default interval:** 60 seconds
- **Configurable range:** 30 seconds to 3600 seconds
- **Staleness guard:** Skips collection if the most recent sample is less than 10 minutes old
- **Concurrency guard:** Prevents overlapping collection runs

Manual Collection

Force an immediate telemetry snapshot via the API: `POST /api/metrics/collect`

28. Lifecycle Workspace

The Lifecycle workspace tracks the operational lifecycle of hosts and VMs.

- **Host Maintenance Posture:** Overview of which hosts are in maintenance mode, scheduled for maintenance, or operational.
- **Lifecycle-Oriented Task Tracking:** Tasks related to lifecycle operations (patching, upgrades, decommissioning).
- **Compliance & Drift Signals:** Indicators showing whether infrastructure state matches expected lifecycle plans.
- **Lifecycle Plans:** Persisted plans that define the desired lifecycle state for hosts and VMs.
- **Remediation Guidance:** Actionable steps to bring drift back into compliance.

29. Capacity Planning

The Capacity workspace provides resource utilization analysis and planning tools.

- **Host Memory Pressure:** Visual indication of memory pressure across hosts.
- **Storage Commitment:** Ratio of allocated to physical storage.
- **Top VM Consumers:** Ranking of VMs by resource consumption (CPU, memory, storage).
- **Noisy-Neighbor Detection:** Identifies VMs disproportionately consuming shared resources.
- **Persisted Telemetry Trends:** Historical trends for capacity forecasting.
- **Background Tasks:** Ongoing operations that affect capacity (migrations, deployments).
- **Operator Guidance:** Recommendations for capacity optimization.

30. Resilience & HA/DR

The Resilience workspace manages High Availability and Disaster Recovery configuration.

Protection Coverage

Overview of which VMs and hosts have HA protection enabled, and which do not.

Failover Readiness

Assessment of the pool's ability to recover from host failures, including available failover capacity.

Evacuation Targets

For each host, identifies which other hosts can receive its VMs in case of failure.

Recovery Runbooks

Create and manage runbooks that define step-by-step recovery procedures:

- **Create Runbook:** Define recovery steps, target objects, and expected outcomes.
- **Execute Drill:** Run a recovery drill (simulation) to validate the runbook.
- **Audit Coverage:** All drill executions are recorded in the audit trail.

31. Inventory & Search

The Inventory workspace provides a universal search across all infrastructure objects.

Universal Search

Search across pools, hosts, VMs, storage repositories, networks, templates, and more using a single search bar. Results are categorized by object type.

Workspace Presets

Save custom inventory views with specific filter criteria and target bindings:

- Create a preset that filters for specific VMs, hosts, or other objects.
- Bind the preset to a specific connection target.
- Quickly switch between saved views.

Saved Connections

Manage all saved XenServer connection targets from the Inventory workspace.

32. Remediation Tasks

Remediation tasks track the resolution of operational issues identified through alerts, lifecycle drift, or manual creation.

Task Lifecycle

1. **Created:** Task is created from an alert, manually, or from a template.
2. **In Progress:** Work is being done to resolve the issue.
3. **Resolved:** The issue has been addressed.
4. **Closed:** The task is archived and subject to retention policies.

Remediation Templates

Create reusable templates for common remediation workflows:

- Define standard steps for recurring issues (e.g., "Host Memory High" → steps to rebalance VMs).
- Templates include recurrence guards to prevent duplicate task creation.
- Templates can be updated and versioned.
- Template deletion requires governance approval.

Part VI — Administration

33. User Management

XenMange manages its own local user accounts independently from XenServer. Users are stored in the `security.db` database.

Creating Users

1. Navigate to **Governance** → **User Management**.
2. Click **Add User**.
3. Enter:
   - **Username:** Unique login name.
   - **Display Name:** Human-readable name.
   - **Email:** Optional contact email.
   - **Password:** Initial password (stored as bcrypt hash).
   - **Role:** `read-only`, `operator`, or `admin`.
4. Click **Create**.

Updating Users

Click **Edit** on any user to modify their display name, email, role, or active status.

Password Rotation

Click **Change Password** on any user to set a new password. Users can also change their own password.

> **Last Admin Protection:** XenMange prevents you from disabling, de-roleing, or deleting the last active admin account. This ensures you cannot lock yourself out of the system.

34. Group Management

Groups allow organizing users for bulk role assignment and resource sharing.

Creating Groups

1. Navigate to **Governance** → **Group Management**.
2. Click **Add Group**.
3. Enter a group name and click **Create**.

Managing Membership

Click **Edit** on a group to add or remove users from the group.

Deleting Groups

Click **Delete** on a group to remove it. Group membership records are also removed.

> **Note:** Group management is restricted to admin-role users only.

35. Role-Based Access Control

XenMange implements three roles with strict permission ordering:

Role	Level	Permissions
read-only	0	View all resources. Cannot perform any write operations, power operations, or administrative actions.
operator	1	All read-only permissions plus: VM power operations, migration, snapshot management, alert triage, remediation tasks. Cannot manage users, groups, or governance policy.
admin	2	Full access to all features: user/group management, governance policy configuration, system settings, credential vault, destructive action approval, and all operator-level actions.

Session Role Switching

Admins can temporarily switch their session role to a lower level (e.g., switch to "operator" to test operator experience). The session role is capped by the user's persisted account role — you cannot elevate beyond your actual permissions.

36. Governance Policy

The governance policy controls how XenMange enforces operational safety.

Configurable Policy Settings

Setting	Description
Default Role	The role assigned to new users by default.
Destructive Action Approval	When enabled, destructive operations (power operations, deletions, retention sweeps) require an approval token from an admin.
Approval TTL	Time-to-live for approval tokens before they expire.

Approval Workflow

1. Operator requests a destructive action (e.g., shutdown a VM).
2. System generates an approval request with a unique token.

3. Admin reviews the request and approves or denies it.
4. If approved, the operator consumes the token to execute the action.
5. The token is invalidated after use or expiration.

All approval decisions are recorded in the audit trail.

Pool Quotas

Configure resource quotas per pool to limit resource consumption:

- **VM Count:** Maximum number of VMs in the pool.
- **Running VM Count:** Maximum simultaneously running VMs.
- **Total Memory:** Maximum aggregate memory allocation.

37. Credential Vault

The Credential Vault securely stores XenServer passwords using AES-256-GCM envelope encryption.

Adding Credentials

1. Navigate to **Settings** → **Credential Vault**.
2. Click **Add Credential**.
3. Enter:
    - **Name:** Descriptive label (e.g., "Production XenServer Root").
    - **Username:** The XenServer username.
    - **Password:** The XenServer password (encrypted before storage).
    - **Target Type:** `pool` or `host`.
    - **Scope:** `private` (visible only to you) or `shared` (visible to other admins).
4. Click **Save**.

Using Stored Credentials

When connecting to a XenServer, select a stored credential instead of entering a password. The password is decrypted server-side only at the moment of XenServer login and is **never** sent to the browser.

Key Rotation

To rotate the vault encryption key:

1. Generate a new 32-byte base64 key.
2. Set `VAULT_ENCRYPTION_KEY_PREVIOUS` to the current key value.
3. Set `VAULT_ENCRYPTION_KEY` to the new key value.
4. Restart XenMange. Existing credentials will be re-wrapped under the new key on next access.
5. Once all credentials have been accessed (and re-wrapped), remove `VAULT_ENCRYPTION_KEY_PREVIOUS`.

> **Important:** In production mode (`NODE_ENV=production`), `VAULT_ENCRYPTION_KEY` is required. If not set, XenMange will refuse to start.

38. Connection & Host Target Management

Connections

Connections are saved XenServer pool connection targets. Manage them via **Settings → Connections** or the API.

- **Add:** Provide host, port, username, and optional stored credential.
- **Set Default:** Mark a connection as the default target.
- **Delete:** Remove a saved connection (requires governance approval if owned).

Host Targets

Individual host targets for standalone hosts not part of a pool connection:

- **Standalone Mode:** Direct host connection without pool context.
- **Pool-Member Mode:** Host target associated with an existing pool connection.

Resource Ownership

All saved connections, host targets, and inventory workspaces carry ownership metadata:

- **Private:** Visible only to the creating user.
- **Shared:** Visible to all admin users.
- Admins can see all resources regardless of ownership.
- Deleting owned resources requires governance approval.

39. Settings & System Configuration

The Settings workspace provides system-wide configuration options:

- **App Identity:** Application name and branding.
- **Timezone:** Server-side timezone for log timestamps and scheduling.
- **Proxy Behavior:** HTTP proxy settings for outbound connections.
- **Session Timeout:** Configure session cookie max age.
- **Telemetry Collection:** Enable/disable the background metrics collector and configure collection interval.
- **Credential Vault:** Manage stored credentials (see Section 37).
- **Retention Policies:** Configure data retention periods (see Section 40).
- **Connections:** Manage XenServer connections (see Section 38).

40. Retention Policies

XenMange automatically purges old data based on configurable retention policies.

Default Policies

Domain	Default Retention	Description
`audit-log`	180 days	Operator and configuration change entries
`remediation-tasks`	90 days	Closed remediation queue items
`auth-events`	60 days	Login/logout activity
`template-deployment-runs`	90 days	Completed deployment tracking
`metric-samples`	7 days	Raw telemetry snapshots
`metric-hourly-rollups`	90 days	Aggregated telemetry rollups

Managing Policies

1. Navigate to **Settings** → **Retention**.
2. Adjust the retention period (in days) for each domain.
3. Enable or disable individual policies.
4. Click **Preview** to see how many records would be purged.
5. Click **Run Sweep** to execute an immediate purge (requires governance approval).

Automatic Scheduler

A background scheduler runs retention sweeps on a configurable interval (default: every 24 hours). After each sweep, an optional `VACUUM` is run on the affected databases to reclaim disk space.

41. Audit Trail

The audit trail records all significant operator actions across the system.

What Gets Audited

- User login/logout events
- VM power operations (start, shutdown, reboot, suspend)
- VM migration and duplication
- Snapshot creation, revert, and deletion
- Host maintenance mode changes
- Connection and credential changes
- User and group management actions
- Governance policy changes and approval decisions

- Template governance actions (promote, deprecate)
- Alert state changes
- Remediation task lifecycle events
- Retention sweep executions

Viewing Audit Logs

Navigate to **Activity** and filter by "Audit" source. Entries are displayed in reverse chronological order with diff tracking showing what fields changed.

Part VII — Reference

42. Complete API Reference

Authentication

Method	Endpoint	Description
POST	`/api/auth/login`	Control-plane login (username/password)
POST	`/api/auth/xen-login`	XenServer connection login
POST	`/api/auth/logout`	Terminate session
GET	`/api/auth/status`	Current auth and connection status

Dashboard

Method	Endpoint	Description
GET	`/api/dashboard`	Aggregate pool/host/VM/storage/network summary
GET	`/api/dashboard/messages`	Recent XenServer messages

Virtual Machines

Method	Endpoint	Description
GET	`/api/vms`	List all VMs
GET	`/api/vms/templates`	List templates
GET	`/api/vms/:ref/snapshots`	List VM snapshots
POST	`/api/vms/start`	Start VM
POST	`/api/vms/shutdown`	Shutdown VM (clean/hard)
POST	`/api/vms/reboot`	Reboot VM (clean/hard)
POST	`/api/vms/suspend`	Suspend VM
POST	`/api/vms/resume`	Resume VM
POST	`/api/vms/:ref/duplicate`	Clone or copy VM
POST	`/api/vms/:ref/migrate`	Migrate VM (same-pool or cross-pool)
POST	`/api/vms/:ref/snapshots`	Create snapshot/checkpoint
POST	`/api/vms/:ref/snapshots/:snap/revert`	Revert to snapshot
DELETE	`/api/vms/:ref/snapshots/:snap`	Delete snapshot
GET	`/api/vms/:ref/export`	Export VM (XVA/metadata)
PUT	`/api/vms/import`	Import VM (XVA/metadata)

Hosts

Method	Endpoint	Description
GET	`/api/hosts`	List all hosts
GET	`/api/hosts/:ref/metrics`	Live host metrics
POST	`/api/hosts/:ref/maintenance/enter`	Enter maintenance mode
POST	`/api/hosts/:ref/maintenance/exit`	Exit maintenance mode
POST	`/api/hosts/:ref/reboot`	Reboot host
POST	`/api/hosts/:ref/shutdown`	Shutdown host

Storage & Networking

Method	Endpoint	Description
GET	`/api/storage`	List all storage repositories
GET	`/api/networks`	List all virtual networks
GET	`/api/pools`	List all pools

Alerts

Method	Endpoint	Description
GET	`/api/alerts`	List all alerts
POST	`/api/alerts`	Create alert
PUT	`/api/alerts/:id`	Update alert state
DELETE	`/api/alerts/:id`	Delete alert
POST	`/api/alerts/bulk-state`	Bulk update alert states
GET/POST/PUT/DELETE	`/api/alerts/policies/*`	Alert policy CRUD

Metrics

Method	Endpoint	Description
GET	`/api/metrics/cluster`	Cluster-wide telemetry history
GET	`/api/metrics/hosts/:ref`	Host telemetry history
GET	`/api/metrics/vms/:ref`	VM telemetry history
GET	`/api/metrics/storage/:ref`	Storage telemetry history
POST	`/api/metrics/collect`	Force telemetry snapshot

Logs & Audit

Method	Endpoint	Description
GET	`/api/logs`	Federated centralized logs
POST	`/api/logs/export`	Export logs (JSON/HTML/PDF)
GET	`/api/audit`	Audit trail entries

Governance & Administration

Method	Endpoint	Description
GET/PUT	`/api/governance`	Governance policy
GET/POST/PUT	`/api/users`	User management
POST	`/api/users/:id/password`	Rotate user password
GET/POST/PUT/DELETE	`/api/groups`	Group management
GET/POST/DELETE	`/api/connections`	Connection management
POST	`/api/connections/:id/default`	Set default connection
GET/POST/PUT/DELETE	`/api/host-targets`	Host target management
GET/POST/PUT/DELETE	`/api/workspaces`	Inventory workspace presets

Settings & Credentials

Method	Endpoint	Description
GET	`/api/settings`	Runtime configuration
GET	`/api/settings/retention/preview`	Retention sweep preview
POST	`/api/settings/retention/run`	Manual retention sweep
GET/POST/PUT/DELETE	`/api/credentials`	Credential vault CRUD

Tasks & Resilience

Method	Endpoint	Description
GET	`/api/tasks`	Recent task history
GET	`/api/resilience`	HA/DR overview
GET/POST/PUT/DELETE	`/api/lifecycle/*`	Lifecycle plan management

43. Environment Variables

See Section 10 for the complete environment variable reference.

44. Default Ports & Paths

Resource	Default	Configurable Via
HTTP Server	Port 3000	`PORT` env var
XenServer API	Port 443 (HTTPS)	Per-connection configuration
Main Database	`./data/xenmange.db`	`DB_PATH` env var
Security Database	`./data/security.db`	`SECURITY_DB_PATH` env var
Vault Database	`./data/vault.db`	`VAULT_DB_PATH` env var
Performance Database	`./data/perf.db`	`PERF_DB_PATH` env var
Client Assets	`/assets/*`	Hardcoded in server
Compiled Client	`/dist/*`	Hardcoded in server

45. Telemetry Alert Thresholds

Entity	Metric	Warning	Critical
Host	`memory_used_percent`	85%	95%
Storage (SR)	`utilization_percent`	80%	92%

Part VIII — Operations

46. Backup & Recovery

What to Back Up

XenMange stores all data in four SQLite files under the `data/` directory:

- `xenmange.db` — Connections, settings, deployment runs
- `security.db` — Users, groups, sessions, auth events, vault key material
- `vault.db` — Encrypted credentials
- `perf.db` — Telemetry data (can be regenerated if lost)

Backup Procedure

1. Stop the XenMange server.
2. Copy all four database files and the `.env` file.
3. Restart the server.

```
# Example backup script
systemctl stop xenmange
cp data/*.db /backup/xenmange/$(date +%Y%m%d)/
cp .env /backup/xenmange/$(date +%Y%m%d)/
systemctl start xenmange
```

> **Note:** The `perf.db` database contains telemetry data that will be regenerated by the metrics collector. If backup size is a concern, you can exclude it.

Recovery

1. Stop the XenMange server.
2. Restore the database files to the `data/` directory.
3. Restore the `.env` file.
4. Start the server.

> **Important:** The `VAULT_ENCRYPTION_KEY` in `.env` must match the key used when the credentials were encrypted. If you lose this key, encrypted credentials in the vault cannot be decrypted.

47. Upgrading

1. Back up all databases (see Section 46).
2. Stop the server.

3. Pull the latest code: `git pull`
4. Install any new dependencies: `npm install`
5. Rebuild the client: `npm run build:client`
6. Start the server: `npm start`

Database schema migrations are handled automatically at startup. No manual migration steps are required.

48. Troubleshooting

Common Issues

Symptom	Likely Cause	Solution
Cannot connect to XenServer	Network connectivity, TLS certificate, or credential issue	Verify network access to port 443. Check that the XenServer hostname/IP is correct. Note: XenMange accepts self-signed certificates by default.
"No active XenServer connection" error	No XenServer session attached	Use the target selector to connect to a XenServer, or go to Settings → Connections to add one.
Dashboard shows empty data	XenServer connection not active or permissions issue	Ensure the connected user has sufficient XenServer privileges (typically root).
Login fails immediately after install	Bootstrap credentials mismatch	Verify `XENMANGE_BOOTSTRAP_USERNAME` and `XENMANGE_BOOTSTRAP_PASSWORD` in your `.env` file.
"Too many login attempts" error	Rate limit triggered	Wait 15 minutes and try again. The rate limit is 20 attempts per 15 minutes per IP.
Server fails to start in production mode	`VAULT_ENCRYPTION_KEY` not set	Generate a 32-byte base64 key and set it as `VAULT_ENCRYPTION_KEY` in your `.env` file.
`better-sqlite3` installation fails	Missing C++ build tools	Install build tools: `sudo apt-get install build-essential python3` (Debian/Ubuntu) or `sudo yum groupinstall "Development Tools"` (RHEL/CentOS).
Metrics not collecting	Telemetry collector disabled or no active connections	Check Settings → Telemetry to ensure collection is enabled. Ensure at least one XenServer connection is active.
Session expires too quickly	Short session max age or clock skew	Adjust `SESSION_MAX_AGE` in the environment configuration. Default is 24 hours.
PDF export fails	Memory or disk space issue	Check available disk space. For very large log exports, the PDF generation may require significant memory.

Log Files

XenMange logs to stdout/stderr. When running under systemd, view logs with:

```
journalctl -u xenmange -f
```

When running directly, logs are printed to the terminal where the server is running.

Database Inspection

To inspect SQLite databases directly, use the `sqlite3` CLI tool:

```
sqlite3 data/xenmange.db ".tables"           # List tables
sqlite3 data/xenmange.db "SELECT * FROM settings LIMIT 10;"  # Query data
```

49. Frequently Asked Questions

Q: Can I run XenMange on multiple servers simultaneously?

A: No. XenMange is designed as a single-process, single-server application. Running multiple instances against the same SQLite databases will cause data corruption. If you need high availability, place XenMange behind a reverse proxy and use process supervision (e.g., systemd, PM2) to auto-restart on failure.

Q: Does XenMange support XenServer 6.x?

A: XenMange targets XenServer 7.x+ and XCP-ng 8.x+. Older versions may work but are not officially supported.

Q: Can I use XenMange with XCP-ng?

A: Yes. XCP-ng is API-compatible with XenServer, and XenMange communicates via the standard XenAPI JSON-RPC protocol.

Q: How do I change the server port?

A: Set the `PORT` environment variable in your `.env` file to the desired port number, then restart the server.

Q: Is TLS required for the XenMange web interface?

A: XenMange serves HTTP by default. For production use, place a reverse proxy (nginx, Caddy) in front to handle TLS termination. The XenMange server itself does not support TLS natively.

Q: How much disk space do the databases require?

A: This depends on the number of VMs, hosts, and the telemetry retention configuration. The main databases are typically small (a few MB). The `perf.db` database grows based on the number of entities and retention period — with default settings (7 days raw, 90 days hourly), expect roughly 10-50 MB for a medium-sized environment (50-200 VMs).

Q: Can I back up just the credential vault?

A: Yes, back up both `vault.db` (encrypted passwords) and the `vault_key_material` table in `security.db` (wrapped DEKs). You also need the `VAULT_ENCRYPTION_KEY` from your `.env` file to decrypt.

Q: How do I completely reset XenMange?

A: Stop the server, delete all files in the `data/` directory, and restart. XenMange will recreate the databases and bootstrap user on next start. Note: this will lose all saved connections, users, credentials, and telemetry data.

Q: What happens if the vault encryption key is lost?

A: Any credentials encrypted with that key cannot be decrypted. You will need to re-enter all XenServer passwords. The application itself remains functional — only stored credentials are affected.

Q: Does XenMange work offline?

A: The web interface loads entirely from local assets (no CDN). However, all infrastructure management features require network connectivity to XenServer hosts.

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